Robot kinematics report

10/18/2026
Robot: **stanford** · 6 DOF · RRPRRR · Craig-style Modified DH

1. Modified DH parameters

$$ {}^{i-1}T_i = Rx(\alpha_{i-1}) \cdot Tx(a_{i-1}) \cdot Rz(\theta_i) \cdot Tz(d_i) $$

i	Joint	Type	α_{i-1} (°)	a_{i-1} (mm)	θ_i (°)	d_i (mm)	Limits
1	Waist	R	0.000	0.000	**0.000***	412.000	-170 … 170°
2	Shoulder	R	-90.000	0.000	**-135.000***	154.000	-135 … 135°
3	Telescope	P	90.000	0.000	0.000	**600.000***	100 … 600 mm
4	Roll 1	R	0.000	0.000	**-86.500***	0.000	-170 … 170°
5	Bend	R	-90.000	0.000	**-37.800***	0.000	-90 … 90°
6	Roll 2	R	90.000	0.000	**69.200***	120.000	-170 … 170°

* joint variable. Base height 140 mm.

2. Worked derivation

Each link transform is built from the table row, then the transforms are multiplied left to right.

$^{i-1}T_i$ for i = 1 (Waist)

$$ Rx(0.000°) \cdot Tx(0.000) \cdot Rz(0.000°) \cdot Tz(412.000) $$

$$ \begin{bmatrix} 1.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 1.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 1.000 & 412.000 \\ 0.000 & 0.000 & 0.000 & 1.000 \end{bmatrix} $$

$^{i-1}T_i$ for i = 2 (Shoulder)

$$ Rx(-90.000°) \cdot Tx(0.000) \cdot Rz(-135.000°) \cdot Tz(154.000) $$

$$ \begin{bmatrix} -0.707 & 0.707 & 0.000 & 0.000 \\ 0.000 & 0.000 & 1.000 & 154.000 \\ 0.707 & 0.707 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 1.000 \end{bmatrix} $$

$^{i-1}T_i$ for i = 3 (Telescope)

$$ Rx(90.000°) \cdot Tx(0.000) \cdot Rz(0.000°) \cdot Tz(600.000) $$

$$ \begin{bmatrix} 1.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & -1.000 & -600.000 \\ 0.000 & 1.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 1.000 \end{bmatrix} $$

$^{i-1}T_i$ for i = 4 (Roll 1)

$$ Rx(0.000°) \cdot Tx(0.000) \cdot Rz(-86.500°) \cdot Tz(0.000) $$

$$ \begin{bmatrix} 0.061 & 0.998 & 0.000 & 0.000 \\ -0.998 & 0.061 & 0.000 & 0.000 \\ 0.000 & 0.000 & 1.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 1.000 \end{bmatrix} $$

$^{i-1}T_i$ for i = 5 (Bend)

$$ Rx(-90.000°) \cdot Tx(0.000) \cdot Rz(-37.800°) \cdot Tz(0.000) $$

$$ \begin{bmatrix} 0.790 & 0.613 & 0.000 & 0.000 \\ 0.000 & 0.000 & 1.000 & 0.000 \\ 0.613 & -0.790 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 1.000 \end{bmatrix} $$

$^{i-1}T_i$ for i = 6 (Roll 2)

$$ Rx(90.000°) \cdot Tx(0.000) \cdot Rz(69.200°) \cdot Tz(120.000) $$

0.355	-0.935	0.000	0.000
0.000	0.000	-1.000	-120.000
0.935	0.355	0.000	0.000
0.000	0.000	0.000	1.000

Final transform ^0T_n = 0T1 · 1T2 · 2T3 · 3T4 · 4T5 · 5T6

-0.826	0.186	-0.532	-488.136
-0.223	0.759	0.612	227.412
0.518	0.624	-0.585	-82.486
0.000	0.000	0.000	1.000

Tool position: X = -488.136 mm, Y = 227.412 mm, Z = -82.486 mm.

3. Jacobian at this pose

Rows: Vx, Vy, Vz (mm/rad or mm/mm), ωx, ωy, ωz (rad/rad).

-227.412	-494.486	-0.707	51.910	-56.100	0.000
-488.136	0.000	0.000	-4.490	-94.642	0.000
0.000	488.136	-0.707	-51.910	-47.914	0.000
0.000	0.000	0.000	-0.707	-0.706	-0.532
0.000	1.000	0.000	0.000	0.061	0.612
1.000	0.000	0.000	-0.707	0.706	-0.585

Singular values (linear part, SI): 1.021, 0.728, 0.468 · manipulability 0.3480 · regular.

4. Robot pose

Figure: stanford at the pose above (joint variables 0.0°, -135.0°, 600.0 mm, -86.5°, -37.8°, 69.2°).

Generated by Crayon Robotics Lab v2.0.0 · 10/18/2026, 5:59:03 PM · engine self-tests 30/30 passed
Reopen this exact robot: file:///F:/store/crayon-robotics-lab-v2/crayon-robotics-lab/index.html#s=eyJqIjpbeyJuYW1lIjoiV2Fpc3QiLCJ0eXBlIjoiUiIsImFscGhhIjowLCJhIjowLCJ0aGV0YSI6MCwiZCI6NDEyLCJtaW4iOi0xNzAsIm1heCI6MTcwfSx7Im5hbWL